

CBD Comparison Data & Result Validity Report

EMS Optimization Project | March 12, 2026

1. Manhattan-Wide vs CBD Comparison Data

1.1 Primary Comparison Table (results/tables/cbd_comparison.csv)

This table directly compares **overall Manhattan** metrics against **CBD-specific** and **non-CBD** subregions under baseline conditions:

Policy	Overall Mean RT	CBD Mean RT	Non-CBD Mean RT	Overall Coverage (8 min)	CBD Coverage	Non-CBD Coverage
P0 (status quo)	8.08 min	2.73 min	14.76 min	64.4%	99.9%	20.3%
P1 (nearest)	2.63 min	2.44 min	2.88 min	99.6%	99.9%	99.1%
P2 (optimized)	2.57 min	2.48 min	2.68 min	99.6%	99.9%	99.2%

Key insight: Under P0, the CBD enjoys excellent service (2.73 min, 99.9% coverage) while non-CBD areas suffer catastrophically (14.76 min, 20.3% coverage). P1 and P2 equalize performance across both subregions.

1.2 Full Scenario Comparison (results/tables/cbd_summary_all.csv)

This table extends the comparison across all 4 CBD stress-test scenarios:

Scenario	Policy	Overall RT	CBD RT	Non-CBD RT	Overall Cov.	CBD Cov.	Non-CBD Cov.	Incidents
baseline	P0	8.08	2.73	14.76	64.4%	99.9%	20.3%	576.9
baseline	P1	2.63	2.44	2.88	99.6%	99.9%	99.1%	574.1
baseline	P2	2.57	2.48	2.68	99.6%	99.9%	99.2%	574.7
cbd_surge (2× demand)	P0	8.35	2.91	15.13	63.1%	99.5%	17.9%	909.6
cbd_surge	P1	2.75	2.53	3.02	99.2%	99.8%	98.4%	907.8
cbd_surge	P2	2.73	2.61	2.87	99.3%	99.8%	98.7%	906.8
cbd_slow_service (35 min)	P0	8.15	2.77	14.88	64.2%	99.9%	19.6%	576.6
cbd_slow_service	P1	2.68	2.47	2.94	99.5%	99.9%	98.9%	574.0
cbd_slow_service	P2	2.63	2.53	2.75	99.6%	99.9%	99.2%	574.6
cbd_only	CBD_ONLY	8.35	2.97	15.05	63.5%	99.9%	18.1%	576.9
mixed	MIXED	8.35	2.97	15.05	63.5%	99.9%	18.1%	576.9

1.3 Figures Showing the Comparison

Three figures visualize the CBD vs Manhattan comparison:

- `results/figures/fig_cbd_comparison.png` — Side-by-side bar chart of CBD vs non-CBD response times
- `results/figures/cbd_response_comparison.png` — Response time distributions by subregion
- `results/figures/cbd_scenario_comparison.png` — Performance across all CBD stress scenarios

- `results/figures/cbd_heatmap.png` — Spatial heatmap of CBD performance

1.4 Documentation

- `docs/cbd_robustness_analysis.pdf` — Full written analysis with interpretation
- `docs/cbd_definition.pdf` — Geographic definition of the CBD (10 precincts with $\geq 30\%$ overlap with MTA Congestion Relief Zone)

2. Validity of Previous Results (Phases 1-7)

2.1 Production Runs: UNCHANGED

The original 1,440 production simulation runs are **completely intact** in their original location:

Experiment	File	Rows	Description
exp1	<code>production/exp1_policy_comparison.csv</code>	90	Policy comparison (P0/P1/P2 × 30 reps)
exp2	<code>production/exp2_fleet_sensitivity.csv</code>	540	Fleet size sensitivity
exp3	<code>production/exp3_demand_sensitivity.csv</code>	540	Demand multiplier sensitivity
exp4	<code>production/exp4_service_robustness.csv</code>	270	Service time robustness
Total		1,440	

Timestamp verification: All production files were last modified at ~20:51-20:54 UTC on March 12, 2026 (the original production run time). They have **not been touched** since.

2.2 CBD Experiment: ADDITIVE

The CBD experiment (330 runs) is stored in a **completely separate directory**:

```
results/simulation/
├── production/           ← Original 1,440 runs (UNTOUCHED)
│   ├── exp1_policy_comparison.csv
│   ├── exp2_fleet_sensitivity.csv
│   ├── exp3_demand_sensitivity.csv
│   └── exp4_service_robustness.csv
└── cbd_experiment/       ← NEW 330 runs (ADDITIVE)
    ├── cbd_experiment_results.csv
    └── cbd_experiment_summary.csv
```

- CBD files were created at 22:10 UTC — **~1.5 hours after** the production runs

- No production files were modified during or after the CBD experiment
- Total project runs: $1,440 + 330 = \mathbf{1,770 \text{ runs}}$

2.3 Queue Analysis: Based on SAME Data

The queue analysis (`results/tables/queue_statistics.csv`) references both production and CBD experiment data. It **reads** from the existing CSVs without modifying them. Key finding confirmed across all 1,770 runs: **queue_fraction = 0.0** everywhere.

2.4 Seasonal Analysis: Based on SAME Underlying Data

The seasonal analysis (`results/tables/seasonal_analysis.csv`) is based on the **original 416,434 crash records** — the same demand dataset used throughout the project. It performs a temporal decomposition of the input data, not the simulation outputs.

2.5 All Previous Deliverables: VALID

Category	Status	Notes
Experiment results (exp1-exp4)	 Unchanged	1,440 runs in <code>production/</code> directory
Statistical tables (ANOVA, CIs, effect sizes)	 Valid	Based on unchanged production data
Publication figures (pub_fig1-5)	 Valid	Generated from unchanged data
Optimization results	 Valid	Based on unchanged simulation outputs
Technical documentation	 Valid	Enhanced with new sections, originals preserved

3. Summary

1. **The CBD vs Manhattan comparison data** is in `results/tables/cbd_comparison.csv` and `results/tables/cbd_summary_all.csv` , with visualizations in `results/figures/cbd_*.png` and full analysis in `docs/cbd_robustness_analysis.pdf` .
2. **All previous results remain 100% valid.** The gap resolution was purely additive — 330 new CBD runs in a separate directory, plus new analysis scripts and documentation. The original 1,440 production runs, all figures, tables, and findings are untouched and unchanged.